

1. What is the truth value of the statement " $P \vee \neg P$ " (P OR NOT P)?
 - a) True
 - b) False
 - c) It depends on P
 - d) None of the above
2. Which of the following is the negation of the statement " $P \wedge Q$ " (P AND Q)?
 - a) $\neg P \vee \neg Q$
 - b) $\neg P \wedge \neg Q$
 - c) $P \vee Q$
 - d) $P \wedge \neg Q$
3. Which of the following is an example of tautology?
 - a) $P \wedge \neg P$
 - b) $P \vee P$
 - c) $P \vee \neg P$
 - d) $P \wedge P$
4. If P is true and Q is false, what is the truth value of the statement $P \rightarrow Q$ (P implies Q)?
 - a) True
 - b) False
 - c) Undetermined
 - d) True if P is true
5. What is the conjunction of the statements "It is raining" (P) and "It is cold" (Q)?
 - a) $P \vee Q$
 - b) $P \wedge Q$
 - c) $P \rightarrow Q$
 - d) $P \leftrightarrow Q$
6. In a conditional statement $P \rightarrow Q$, which of the following is the contrapositive?
 - a) $\neg P \rightarrow \neg Q$
 - b) $Q \rightarrow P$
 - c) $\neg Q \rightarrow \neg P$
 - d) $P \rightarrow Q$
7. Which of the following is true about the statement "If P, then Q"?
 - a) P is the consequent, and Q is the antecedent
 - b) P is the antecedent, and Q is the consequent
 - c) P and Q must both be false for the statement to be true
 - d) The truth of P guarantees the falsity of Q
8. The converse of the implication $P \rightarrow Q$ is
 - a) $\neg Q \rightarrow \neg P$
 - b) $P \vee \neg Q$
 - c) $Q \rightarrow P$
 - d) $P \wedge Q$
9. The statement "If it is raining, then the ground is wet" is equivalent to which of the following?
 - a) If the ground is not wet, it is not raining
 - b) If it is not raining, the ground is wet
 - c) The ground is wet if and only if it is raining
 - d) If the ground is wet, it is raining
10. Which of the following is NOT a logical equivalent of the statement $P \rightarrow Q$?
 - a) $\neg P \vee Q$
 - b) $\neg Q \rightarrow \neg P$
 - c) $Q \vee P$
 - d) $\neg Q \wedge P$
11. Which of the following is a valid argument form?
 - a) Affirming the antecedent
 - b) Denying the antecedent
 - c) Affirming the consequent
 - d) Denying the consequent

12. What type of reasoning is used when the conclusion necessarily follows from the premises?
- a) Inductive b) Deductive c) Hypothetical d) Circular
13. If a logical argument is valid, then
- a) The conclusion must be true b) The premises must be false
c) The conclusion follows from the premises d) The argument is sound
14. The fallacy of affirming the consequent occurs when
- a) $P \rightarrow Q$ is true and Q is true, therefore P is true
b) $P \rightarrow Q$ is true and P is false, therefore Q is false
c) $P \rightarrow Q$ is true and Q is false, therefore P is false
d) $P \rightarrow Q$ is true and P is true, therefore Q is true
15. An argument with true premises and a false conclusion is always
- a) Valid b) Invalid c) Sound d) None of the above
16. The expression $\neg(P \wedge Q)$ is logically equivalent to
- a) $P \vee Q$ b) $\neg P \vee \neg Q$ c) $P \wedge \neg Q$ d) $\neg P \wedge Q$
17. The statement $P \rightarrow Q$ is logically equivalent to
- a) $\neg P \rightarrow \neg Q$ b) $\neg P \vee Q$ c) $P \wedge \neg Q$ d) $P \vee \neg Q$
18. Which of the following is a De Morgan's Law?
- a) $\neg(P \vee Q) = \neg P \wedge \neg Q$ b) $\neg(P \wedge Q) = P \vee \neg Q$
c) $\neg(P \vee Q) = \neg P \vee Q$ d) $P \vee (Q \wedge R) = (P \vee Q) \wedge (P \vee R)$
19. Which of the following is NOT a logical equivalence?
- a) $P \wedge P = P$ b) $P \vee P = P$ c) $P \wedge Q = Q \wedge P$ d) $P \wedge (P \vee Q) = P$
20. The expression $(P \wedge Q) \vee (P \wedge \neg Q)$ is logically equivalent to
- a) P b) Q c) $P \wedge Q$ d) $\neg P$
21. Which logic gate gives an output of 1 only if all inputs are 1?
- a) AND gate b) OR gate c) NOT gate d) XOR gate
22. What is the output of an XOR gate when both inputs are 1?
- a) 1 b) 0 c) Undefined d) Same as input
23. The truth table for a NOT gate has
- a) Two input and two output combinations b) One input and two output combinations
c) Two input and one output combination d) One input and one output combination
24. Which gate is called a universal gate because it can be used to construct any other logic gate?

